

SymPy Bug Card

Bug 12: inverse Fourier transform drops $|t|$

Task. Compute the inverse Fourier transform of the Lorentzian

$$F(w) = \frac{2}{1 + 4\pi^2 w^2}.$$

SymPy's answer.

$$\text{inverse_fourier_transform}(F(w), w, t) = e^{-t}.$$

Correct answer.

$$\text{inverse_fourier_transform}(F(w), w, t) = e^{-|t|}.$$

Explanation. The input $F(w)$ is real and even in w , so its inverse Fourier transform should be real and even in t . SymPy returns e^{-t} , which is not even: for example, at $t = -1$ it gives e , while the expected value is e^{-1} . The missing absolute value changes the answer on the negative half-line.

Likely root cause. In `sympy/integrals/transforms.py`, `_fourier_transform` appears to select the first branch of a conditional `Piecewise` integral, then the public inverse transform drops the condition under the default `noconds=True`. This is a new instance of a known failure mode.

Artifact (GitHub). [bug-report/bug-012-inverse-fourier-transform-drops-abs-t/](#)